LARAM School 2007: 1st week

	8.30	9.30	10.30	11.30	12.30	13.30	14.30	15.30	16.30	17.30
Mon 03/09		I-1 Cascini	I-2 Corominas	break	I-2 Corominas	Lunch		I-3 Cascini	break	I-3 Cascini	I-4 Vardoul.
Tues 04/09	I-4 Vardoulakis		break	II-1 Nadim		Lunch		II-2 Corominas	break	II-3 Nadim	
Wed 05/09	II-4 Corominas		break	II-5 (from 11.15) Nadim		Lunch		II-6 Pastor	break	II-7 Pastor	
Thur 06/09	II-8 Pastor		break	II-9 Cor., Bonn., Nadim		Lunch		II-10 (until 15.45) Bonnard	break	II-11 students	
Fri 07/09	II-12 Bonnard		break	II-13 Bonnard		Lunch	III-1 Sor.	III-2 Bonnard	break	III-2 Bonnard	III-3 Sorbino
Sat 08/09	Technical VISIT										

Session I - "Introduction to landslides" & "Landslide analysis using approaches based on: Geology, Geotechnics and Geomechanics"

I-1 (Cascini, 1h): Introduction to landslides

I-2 (Corominas, 2.5hs): Landslide identification. Key geological, geomorphological and hydrogeologicla features of landslides

I-3 (Cascini, 2hs): The geotechnical slope model

I-4 (Vardoulakis, 3hs): Basic geomechanics of landslides: Graphostatics, the action of water and dynamics

Session II - "Risk Theory and Risk Analysis for Landslides"

II-1 (Nadim, 2hs): Introduction to risk theory and risk analysis for landslides: what is risk, problems to solved at different scales for quantification of risk for different categories of landslides

II-2 (Corominas, 1.5hs): Empirical models for travel distance estimation

II-3 (Nadim, 1.5hs): Deterministic and probabilistic models for slope stability evaluation

II-4 (Corominas, 2hs): Landslide frequency assessment

II-5 (Nadim, 1.75hs): Application examples of probabilistic methods and semi-quantitative methods for landslide hazard zonation

II-6 (Pastor, 1.5hs): Introduction to modelling of catastrophic landslide events

II-7 (Pastor, 1.5hs): Advanced numerical models for initiation of landslides

II-8 (Pastor, 2hs): Advanced numerical models for propagation of sediments (with emphasis on application examples and comparison with empirical model predictions)

II-9 (Corominas, Bonnard, Nadim, 2hs): Case studies: Andorra rockfalls (Corominas), Sedrun village in Switzerland (Bonnard), Åknes rock slope in Norway (Nadim)

II-10 (Bonnard, 1.25hs): Different components of vulnerability to landslides (physical, social, economical, environmental).

II-11 (students, 1.5hs): Posters by students

II-12 (Bonnard, 2hs): Prevention and long-term management of landslides, example from motorway in La Réunion Island

II-13 (Bonnard, 1.5hs): Climate change and its potential effects on the landslide hazard in Europe

Session III - "Landslide risk mitigation by control works"

III-1 (Sorbino, 0.5h): Control works according to the risk mitigation framework

III-2 (Bonnard, 2hs): Principles of prevention and long-term management of landslides, efficiency of drainage works

III-3 (Sorbino, 1h): Case histories

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LARAM School 2007: 2nd week

	8.30	9.30	10.30	11.30	12.30	13.30	14.30	15.30	16.30	17.30
Mon 10/09	IV-1 Cascini		break	IV-2 Cascini		IV-3 Cascini	Lunch	IV-4 van Westen	break	IV-5 van Westen
Tue 11/09	IV-6 van Westen			break	IV-7 Ho	Lunch	IV-8 Ho	IV-9 Ho	break	IV-10 Ho
Wed 12/09	IV-11 Ho	V-1 Cascini	break	V-2 Sorbino		Lunch	V-2 Sorbino	break	V-3 Nova	
Thur 13/09	V-3 Nova	V-4 Darve	break	V-4 Darve		Lunch	VI-1 Leroi	break	VI-1 Leroi	
Fri 14/09	VI-1 Leroi		break	VI-2 Lacerda		Lunch	VI-3 Lacerda	break	VI-4 Seco Pinto	Concluding remarks
Sat 15/09	Departures									

Session IV - "Landslide susceptibility, hazard and risk zoning at different scales"

IV-1 (Cascini, 1.5hs): The International "Guidelines for Landslide Susceptibility, Hazard and Risk Zoning for Land Use Planning". Input elements to zoning maps and Zoning at different scales.
IV-2 (Cascini, 1.5hs): Basic methods and procedures for susceptibility and hazard zoning at small and medium scale (1:100,000-25,000)
IV-3 (Cascini, 1h): Tutorial on susceptibility and hazard zoning at 1:25,000 scale
IV-4 (van Westen, 1.5hs): GIS for landslide susceptibility and hazard assessment: spatial data
IV-5 (van Westen, 1.5hs): GIS for landslide susceptibility and hazard assessment: spatial analysis
IV-6 (van Westen, 2.5h): Tutorial on landslide susceptibility assessment using statistical analysis
IV-7 (Ho, 1h): Natural terrain zoning and management criteria - Hong Kong practice and experience
IV-8 (Ho, 1h): Qualitative risk rating for individual slopes/hillsides and global quantitative risk assessment
IV-9 (Ho, 1.5hs): Site-specific quantitative risk assessment and risk management
IV-10 (Ho, 1h): Tutorial on quantitative risk assessment
IV-11 (Ho, 1h): Digital technology for mapping and monitoring

Session V - "The role of sophisticated methods in landslide Risk analysis" and "Improvement of sophisticated methods with the aid of geomechanics"

V-1 (Cascini, 1h): Introduction to advanced slope stability characterization
V-2 (Sorbino, 3hs): Flow-like mass movements in pyroclastic soils: Triggering mechanisms and some remarks on propagation stage
V-3 (Nova, 3hs): Analysis of the stability of soil slopes with low slope angles as a result of latent instability
IV-4 (Darve, 3hs): Finite element modelling of landslides by taking into account an hydromechanical coupling and an instability criterion

Session VI - "Landslide risk management"

VI-1 (Leroi, 5hs): Risk management on la Désirade Island and Pointe-Noire in Guadeloupe
VI-2 (Lacerda, 2hs): Engineering solutions to mitigate landslide risk
VI-3 (Lacerda, 1h): Landslide Risk Management in Rio de Janeiro
VI-4 (Seco e Pinto, 1h): Case history: Risk management of a poorly-designed 30 m high slope

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